Catena 243 (2024) 108174

Contents lists available at ScienceDirect

Catena

journal homepage: www.elsevier.com/locate/catena

Distinct variations of soil infiltrability of contrasting root types in a temperate mosaic-pattern grassland in northern China

Peipei Wang [a], Qian Liu [a], Zhengchao Zhou [a,b,*], Jun'e Liu [a,b], Liguo Cao [a,b], Ning Wang [a,b], Yuying Cao [a]

[a] School of Geography and Tourism, Shaanxi Normal University, Xi'an 710119, China
[b] National Demonstration Centre for Experimental Geography Education, Shaanxi Normal University, Xi'an 710119, China

ARTICLE INFO

Keywords:
Soil infiltration capacity
Patchy mosaic pattern
Root traits
Naturally restored grassland
Water-limited area

ABSTRACT

Soil infiltration capacity is an important factor affecting soil moisture, and is of great significance to plant growth and groundwater recharge in arid and semiarid areas. Patchiness of grasslands widely exist and have a crucial effect on hydrological processes in water-limited regions. This study investigated soil infiltrability under natural grasslands with patchy mosaic patterns based on a series of in-situ tests. Two dominated natural restoration vegetation communities (*Bothriochloa ischaemum* (*B. ischaemum*) with fibrous root systems and *Artemisia gmelinii* (*A. gmelinii*) with tap root systems) and abandoned land (as control) were selected in the loess hilly and gully region of the Loess Plateau, China. The results showed that natural restoration grasslands significantly improved soil infiltrability in both plant and bare patches, and the emergence of patchy mosaic patterns can result in infiltration heterogeneity to redistribute water on a hillslope scale. Compared to *B. ischaemum*, *A. gmelinii* are more likely to facilitate the formation of preferential flow, contributing to the increased soil infiltration rates. The initial, steady and average infiltration rates (*IIR*, *SIR*, and *AIR*, respectively) and soil infiltration capacity index (*SIC*) were all negatively correlated with both root length density (*RLD*) and root surface area density (*RSAD*) of fine roots (diameter < 2 mm) but positively correlated with those of coarse roots (diameter ≥ 2 mm). The main factors influencing the infiltration capacity were the mean weight diameter in the 10–20 cm layer and the *RSAD* of the fine roots in the 0–20 cm layer, which can explain 98.6 %, 81.7 %, 63.9 %, and 91.3 % of the total variance of *IIR*, *SIR*, *AIR*, and *SIC*, respectively. This work highlights the important roles of patchy mosaic patterns in affecting soil water redistribution and the stabilization of ecological systems, providing a scientific basis for vegetation selection in the process of vegetation restoration in water-limited regions.

1. Introduction

Infiltration is an important process of hydrological cycle in terrestrial ecosystem, which profoundly affects water budget, soil water replenishment, and surface runoff (Khan et al., 2016; Sun et al., 2018; Zhao et al., 2013). This process is particularly crucial for ecosystems in arid and semi-arid regions, where soil moisture supplying for vegetation survival and growth mainly originates from rainfall infiltration (Guo et al., 2019; Wu et al., 2021). In these water-stressed environments, observed decline in soil moisture availability tends to exacerbate soil water scarcity and consequently threaten the sustainability of vegetation restoration (Jia et al., 2019). In this regard, model simulations have projected that soil moisture variability amplifies greenhouse gas-driven global warming when it changes coupled with atmosphere due to the "warmer climate – drier soil" feedback (Qiao et al., 2023).Therefore, a quantitative assessment of soil infiltrability and its response to environmental changes are urgently required in water-limited regions, which will facilitate an accurate projection of soil moisture contribution to future warming.

Grassland, the most typical vegetation restoration type in arid and semi-arid areas, has good drought resistance and adaptability to ecologically fragile soil conditions (Cui et al., 2018). Grasslands in water-limited regions usually show self-organized spatial patterns, namely a two-phase mosaic of plant patches and neighbouring bare patches, which is a result of the interplay between soil, vegetation, and hydrological processes (Getzin et al., 2016; Li et al., 2022; Zhang and Hu, 2019). This two-phase mosaic is characterized by discontinuous vegetation cover consisting of high-cover patches (plant patches) that

* Corresponding author.
E-mail address: zczhou@snnu.edu.cn (Z. Zhou).

https://doi.org/10.1016/j.catena.2024.108174
Received 7 October 2023; Received in revised form 7 April 2024; Accepted 7 June 2024
Available online 10 June 2024

have vascular plants and relatively low-cover patches (bare patches) that have sparse (extremely lower levels of plants) or absent plants (Aguiar and Sala, 1999; Molaeinasab et al., 2021; Wu et al., 2016). Besides, mosaic landscape is actually a source-sink system that regulates the transference and storage of vital resources such as water and nutrients, with the bare patch acting as a generator of runoff and sediment and the plant patch acting as a sink (Liu et al., 2013; Molaeinasab et al., 2021). In this case, spatial heterogeneity in vegetation cover and resources between contrasting patches lead to distinct behaviours in root traits and soil properties (Bochet et al., 2006; Liu et al., 2013), which subsequently yields a joint influence on soil infiltration capacity (Bodner et al., 2014; Hao et al., 2020). Thus, exploring the effect of mosaic-pattern grassland patches on soil infiltration is important for a comprehensive understanding of coupled processes of soil-vegetation-hydrology in water-limited environments. Previous studies have widely reported that plant root growth or decay affects soil infiltration capacity by altering soil structure associated with the stability of soil aggregates and the size and connectivity of soil porosities (Bronick and Lal, 2005; Freschet and Roumet, 2017). This means that the impacts of plant roots on soil infiltrability highly depend on various root traits and complex root-soil interactions (Archer et al., 2002; Wang et al., 2023). However, existing studies have focused on root-soil interactions and their influences on infiltration capacity in the rhizosphere among different plant species or between natural and artificial grasslands (Huang et al., 2016; Leung et al., 2015), lacking a direct comparative study on soil infiltration capacity variability under grasslands of two-phase mosaic patterns dominated by different root systems (e.g., fibrous root system and tap root system). In fact, fine roots easily blocked water flow paths by clogging soil micropores and eliminating macropores, while coarse roots tend to cause small-scale compaction and macropore development to improve water flow paths and induce the formation of preferential flow (Li et al., 2022; Lu et al., 2020). Therefore, more studies are required to clarify the root-soil interactions in different root system-dominated grasslands and their impacts on soil infiltration under grasslands of two-phase mosaic patterns.

The loess hilly and gully region is not only a typical arid and semi-arid region but also an important area for ecological development in China. Due to the scarcity of water in this region, vegetation generally distributes following a spatially heterogeneous structure with a mosaic pattern. Many studies on soil infiltration in loess hilly regions have primarily focused on different land cover types (Anderson et al., 2020; Wang et al., 2013), vegetation types (Lozano-Baez et al., 2019; Tang et al., 2018) and restoration chronosequences (Gu et al., 2018; Zhao et al., 2013). The understanding of the effects of roots on soil infiltration under two-phase mosaic patterns in naturally revegetated grasslands and the potential mechanisms remain limited. This knowledge gap impedes an accurate evaluation on the effect of this mosaic pattern on ecohydrological processes and restricts ecological development in the loess hilly and gully region.

This study was carried out based on two typical naturally-restored vegetation communities, *Bothriochloa ischaemum* (*B. ischaemum*) with fibrous root systems and *Artemisia gmelinii* (*A. gmelinii*) with tap root systems, and abandoned land in the loess hilly and gully region. We aim to quantitatively analyse the changes in soil infiltration capacity under naturally revegetated grasslands with different root system types, to compare soil infiltration capacity between plant patch and bare patch in the different plant communities, and to explore the effects of root traits and soil properties on soil infiltrability of grasslands under two-phase mosaic patterns. This study hypothesized that i) soil infiltration would be improved during the process of vegetation restoration in both plant and bare patches with ameliorating soil and root traits, ii) patchy mosaic patterns in grasslands would lead to heterogenous infiltration and redistribute water on slope scale, iii) two plant species with contrasting root system types would differently affect soil infiltrability.

2. Materials and methods

2.1. Study area

The experiment was conducted in the Fangta small watershed (Fig. 1, 36°46′13″–36°49′18″N, 109°15′22″–109°17′01″E), located in the hilly and gully region of the Loess Plateau, China. The region is characterized by a semi-arid continental climate, with an annual average temperature of 8.8 ℃ and a mean annual precipitation of 505 mm, 70 % of which occurs from July to September (Wang et al., 2019). The soil texture is silty loam (clay: 6.3 %, silt: 64.4 %, sand: 29.3 %) based on the International Society of Soil Science soil texture classes. There is a significant amount of natural vegetation across the research region, with herbaceous plants making up the majority of them. The main species in the watershed are *Artemisia gmelinii*, *Stipa bungeana*, *Lespedeza davurica*, and *Bothriochloa ischaemum*. Moreover, the natural grasslands usually exhibit mosaic-pattern, including plant patch and bare patch (Fig. 1b), due to water restrictions in this region.

2.2. Experimental design and sampling

Based on Liu et al. (2019a) and field investigation, the distribution of root systems in this research region stabilized after 12 years of natural vegetation restoration. Therefore, two dominant species of naturally restored grasslands (more than 12 years), including *Bothriochloa ischaemum* (*B. ischaemum*) with fibrous root systems and *Artemisia gmelinii* (*A. gmelinii*) with tap root systems, were selected, and abandoned land (farmland left fallow for one year) was used as the control (Table 1). Subsequently, plant patches and bare patches were chosen in each community, and three replicate plots of 2 × 2 m size were randomly established for each patch type. Soil and root sampling were conducted in identical and adjacent areas to the infiltration experiments at 0–50 cm soil depth with 10 cm intervals. At each depth, soil samples were collected using 100 cm^3 (d = 50.46 mm, h = 50 mm) soil cylinders and an aluminium box to measure soil properties, respectively. Soil cubes (10 × 10 × 10 cm) containing roots were sampled within selected plant patch plots to measure root traits. All the field experiments were conducted after a five-day period with no rainfall.

2.3. Soil infiltration measurement

The infiltration processes were evaluated based on the measurement method of Zhang et al. (2017). Firstly, the plants and litter were removed carefully from the soil surface. Subsequently, a customized double-ring infiltrometer (30 cm diameter inner ring, 60 cm diameter outer ring, both 20 cm in height) was placed on the soil surface and was gently and vertically inserted 5 cm into the soil using a rubber hammer while ensuring that the ring remained horizontal during insertion. Vaseline was applied where the inner ring contacted the soil to prevent water from seeping out.

Next, Brilliant Blue FCF solution (4 g/L) (Flury and Flühler, 1995) was applied to the inner ring, while clean water was poured into the outer ring with water to a height of 5 cm at the same time. The falling head method was used to measure the soil infiltration process (Zhang et al., 2019). Thereafter, the water level in the inner ring was recorded every min for the first 30 min, then every 3 min until steady-state infiltration was achieved (constant infiltration for 3 consecutive measurements was considered as stable infiltration). Water of inner and outer ring would be refilled up to the 5 cm height after each recording. The average infiltration rate of last 3 times constant infiltration was considered as the stable infiltration rate (*SIR*), the average infiltration rate of the first three minutes was denoted the initial infiltration rate (*IIR*), and the overall average infiltration rate of entire infiltration time was regarded as average infiltration rate (*AIR*).

Subsequently, the experimental area was covered with plastic film to avoid interference from evaporation, precipitation and other external

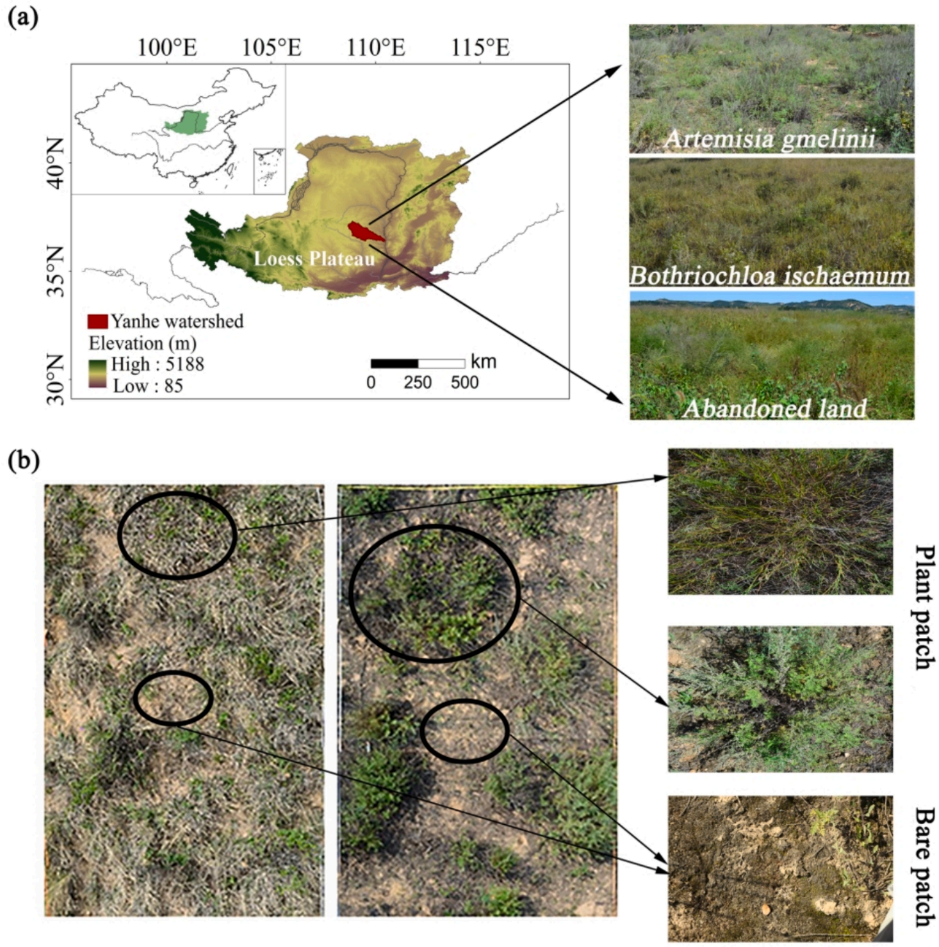

Fig. 1. Overview of the study area and sampling sites. (a) Location of the study area and pictures of the tested sites. (b) The graphs show mosaic-patterned grasslands and patch types (plant patch and bare patch).

Table 1
Basic information on the experimental sites.

Community	Recovery age (a)	Type	Dominant plant species	Latitude (N) Longitude (E)	Slope gradient (°)	Vegetation Coverage (%)
B. ischaemum	18	Perennial herbs with fibrous roots	*Bothriochloa ischaemum*	36°49′2″ 109°16′5″	20	37
A. gmelinii	24	Semi-shrub with tap roots	*Artemisia gmelinii*	36°50′10″ 109°16′19″	18	44
Abandoned land	1	Annual herbaceous plants	*Artemisia scoparia-Setaria viridis*	36°48′39″ 109°15′7″	12	35

factors. Vertical soil profiles were excavated from the centres of the inner rings, 24 h after the infiltration tests were concluded. The depth of the stained soil area was then measured with a steel ruler.

2.4. Soil properties and root traits measurements

The soil bulk density (*BD*, g/cm^3) was measured by oven-drying the samples at 105 ℃ for 24 h, and soil total porosity (*TP*, %) was calculated based on the obtained *BD* data following Eq. (1):

$$TP = (1 - BD/PD) \times 100 \tag{1}$$

PD is the soil particle density (g/cm^3), and *PD* was assumed to be 2.65 g/cm^3.

The soil aggregates were measured using a DIK-2012 soil aggregate analyser, as described by Wang et al. (2023). Briefly, the air-dried bulk soil was first dry sieved and the mass of each fraction was weighed. Then 50 g of soil was weighed for wet sieved according to the mass ratio of dry-sieved aggregates with sieve of 2, 1, 0.5, 0.25, and 0.106 mm. Finally, the aggregates of each size were collected, dried and weighed. As the comprehensive index of the water-stable aggregates, the average weight diameter (*MWD*, mm) was calculated by Eq. (2):

$$MWD = \sum_{i=1}^{i=6} M_i \times D_i / \sum_{i=1}^{i=6} M_i \tag{2}$$

where M_i (g) is the mass of the stable soil aggregates in each diameter class, and D_i (mm) is the mean diameter of each size class.

Following field sampling, the roots were cleaned from free soil through a 0.25 mm sieve and then scanned on a flatbed scanner at 300 dpi (Epson V700 photo scanner, Tokyo, Japan). The root parameters were determined by WinRhizo software (2009a, Regent Instruments, Quebec, QC, Canada). Roots were divided into fine roots (diameter < 2 mm) and coarse roots (diameter ≥ 2 mm) based on the root classification defined by Archer et al. (2002). Subsequently, the roots were weighed after oven drying to a constant weight at 60 °C. Finally, the total root mass density (*RMD*, mg/cm^3) and the root length density (*RLD*, cm/cm^3) and root surface area density (*RSAD*, cm^2/cm^3) of coarse roots (diameter ≥ 2 mm) and fine roots (diameter < 2 mm) were calculated by Eq. (3–5), respectively, as described by Ye et al. (2017) and Zhou and Shangguan (2007).

$$RMD = RM/V \tag{3}$$

$$RLD = RL/V \tag{4}$$

$$RSAD = RSA/V \tag{5}$$

where *RM* (mg) is the total dry root mass, *RL* (cm) and *RSA* (cm^2) denote the root length and the root surface area at each diameter class, and *V* (cm^3) is the volume of the soil cube containing the roots.

2.5. Soil infiltration capacity index

In previous studies, soil infiltration capacity was expressed by infiltration rates in different infiltration states (Fischer et al., 2014; Zhao et al., 2013), particularly in stable and average infiltration states (Liu et al., 2019b; Neris et al., 2012; Reynolds and Reddy, 2012). However, when these parameters are used individually, it is difficult to fully interpret the soil infiltration capacity. Therefore, principal component analysis (PCA) is performed between the *IIR*, *SIR*, and *AIR* to produce an integrated index to comprehensively describe the soil infiltration capacity index (*SIC*), following a similar method as that depicted by Wu et al. (2016). The eigenvector of first principal component (PC1) is 2.638 (>1), explaining 87.94 % of the total variance. The loadings of *IIR*, *SIR*, and *AIR* on PC1 are 0.944, 0.901, and 0.967, respectively. In terms of the scores of each factor in this study, the *SIC* was calculated as follows:

$$SIC = 0.358IIR + 0.341SIR + 0.367AIR \tag{6}$$

2.6. Data analyses

One-way analysis of variance (ANOVA) followed by Duncan's post hoc tests were carried out to detect differences in the soil properties, root traits, and *IIR*, *SIR*, and *AIR* of the different communities. T-tests were conducted to identify the difference in soil properties, *IIR*, *SIR*, and *AIR* between plant patches and bare patches. Pearson correlation analysis was applied to examine the relationships between the infiltration rates, soil properties and root traits. The relative contributions of soil properties and root traits to soil infiltration rates were assessed by using variation partitioning analysis. The main factors affecting the soil infiltration rates were determined by stepwise multiple regression. All the statistical analyses were carried out with R 4.1.2 and IBM SPSS 22.

3. Results

3.1. Soil and root traits under mosaic pattern in different communities

Compared with the abandoned land, the *B. ischaemum* and *A. gmelinii* communities show higher *MWD* and *TP*, especially in the 0–20 cm soil layer (Table 2). The mean *TP* values of the samples from the *B. ischaemum* are much higher than those from the other communities in most layers. However, the highest values of *MWD* appear in the plant patches in the *A. gmelinii* community, followed by those in the *B. ischaemum* community and the abandoned land at a depth of 0–20 cm. Specifically, there is a significant difference ($p < 0.05$) in the *MWD* of the plant patches between the *A. gmelinii* community (1.72–1.89 mm) and the abandoned land (0.36–0.99 mm). With regard to the *MWD* in the bare patches, significant differences are observed between both plant communities and the abandoned land at a depth of 10–20 cm, with a significantly higher *MWD* value (1.49 ± 0.09 mm) in the *B. ischaemum* community than in the other two communities at the 20−30 cm depth ($p < 0.05$). The *MWD* and *TP* in the plant patches are mostly higher than that in the bare patches for each community.

Root mass densities (*RMDs*) decrease with soil depth, and more than 80 % of roots are distributed in the 0–20 cm soil layer (Fig. 2a). The highest *RMD* value is observed in the *A. gmelinii*, followed by the *B. ischaemum* and abandoned land at each soil depth, except for the 0–10 cm layer (Fig. 2a). Moreover, the roots in the *A. gmelinii* consist of 74 % coarse roots (diameter ≥ 2 mm) and 26 % fine roots (diameter < 2 mm), while there are more fine roots in the *B. ischaemum* and abandoned land, accounting for 99 % and 76 % of the total length, respectively (Fig. 2b).

The *B. ischaemum* display the highest values of RLD_{fine} and $RSAD_{fine}$ at all depths, while the *A. gmelinii* exhibit the highest RLD_{coarse} and $RSAD_{coarse}$ (Table 3). Furthermore, the RLD_{fine} of the *B. ischaemum* community is significantly higher than that of the *A. gmelinii* community and abandoned land ($p < 0.05$), and the RLD_{coarse} and $RSAD_{coarse}$ of the *A. gmelinii* community show significant differences with those of the *B. ischaemum* community ($p < 0.05$; Table 3).

3.2. Soil water infiltration under mosaic pattern in different communities

The infiltration rates of the three communities decrease rapidly in the initial state (*IIR*, 0–3 min) and then decrease slowly until they stabilize in the steady state (*SIR*, Fig. 3a). For each community, the infiltration rates in the plant patches is higher than that in the bare patches in the same state, and the stable infiltration state for the bare patches appears earlier than that for the plant patches. Moreover, the infiltration rates and cumulative infiltration exhibit the following order: *A. gmelinii* > *B. ischaemum* > abandoned land (Fig. 3a).

Higher infiltration rates (*IIR*, *SIR*, and *AIR*) are identified for the *A. gmelinii* and *B. ischaemum* communities than for the abandoned land

Table 2
Soil physical properties (mean ± SD) in the 0 − 30 cm soil layer in the different communities.

Communities	Soil depth(cm)	*MWD* (mm)		*BD* (g/cm^3)		*TP* (%)	
		Plant patch	Bare patch	Plant patch	Bare patch	Plant patch	Bare patch
Artemisia gmelinii	0–10	1.89 ± 0.27a	0.84 ± 0.11a	1.17 ± 0.07a	1.18 ± 0.09ab	55.66 ± 2.39b	55.48 ± 3.18ab
	10–20	1.72 ± 0.29a	1.19 ± 0.30a	1.24 ± 0.04a	1.22 ± 0.09a	53.09 ± 1.93a	53.91 ± 2.78a
	20–30	0.93 ± 0.12a	0.40 ± 0.03b	1.28 ± 0.11a	1.29 ± 0.11a	51.87 ± 3.26a	51.46 ± 0.34a
Bothriochloa ischaemum	0–10	1.45 ± 0.33ab	1.35 ± 0.64a	1.05 ± 0.13b	1.15 ± 0.13b	60.25 ± 0.93a	56.58 ± 2.10a
	10–20	1.37 ± 0.14a	1.37 ± 0.11a	1.21 ± 0.06a	1.22 ± 0.06a	54.43 ± 0.87a	53.97 ± 0.90a
	20–30	0.99 ± 0.42a	1.49 ± 0.09a	1.24 ± 0.05a	1.29 ± 0.12a	53.11 ± 0.43a	51.46 ± 1.41a
Abandoned land	0–10	0.99 ± 0.51b	0.73 ± 0.43a*	1.16 ± 0.02a	1.29 ± 0.05a*	56.08 ± 0.71b	51.22 ± 1.87b*
	10–20	0.36 ± 0.08b	0.68 ± 0.08b*	1.26 ± 0.07a	1.20 ± 0.04a	52.62 ± 2.78a	54.55 ± 1.63a
	20–30	0.76 ± 0.06a	0.57 ± 0.18b	1.35 ± 0.04a	1.42 ± 0.22a	49.14 ± 1.41a	46.26 ± 8.20a

Note: *MWD*, mean weight diameter of the soil aggregates; *BD*, bulk density; *TP*, total porosity.
Different letters indicate significant differences among the different vegetation communities within the same patch type at a significance level of $p < 0.05$ (Duncan's post hoc tests). * indicates significant differences between different patches within the same community at a significance level of $p < 0.05$.

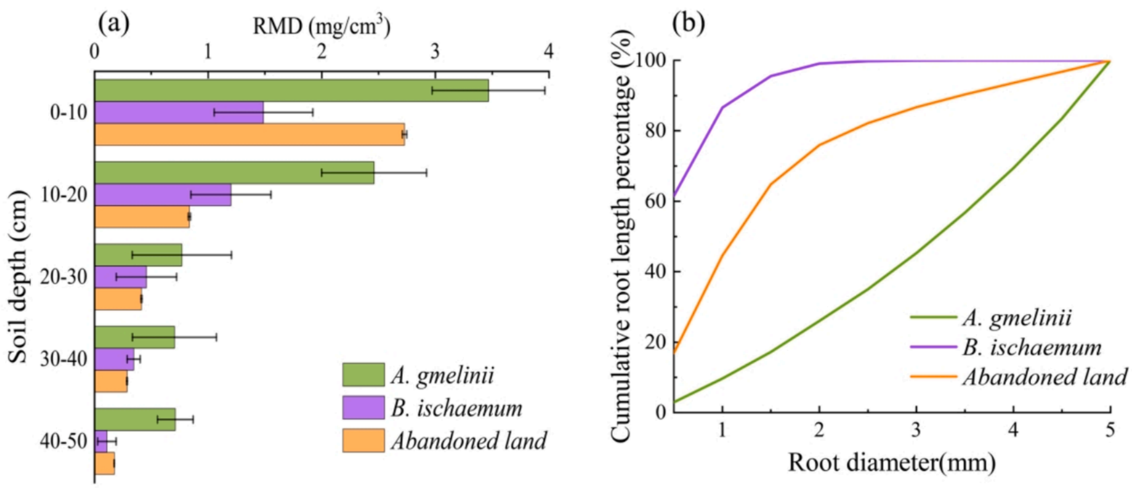

Fig. 2. Root traits of different species. (a) Root mass densities (*RMD*) of different species at a soil depth of 0–50 cm. (b) Cumulative length percentage of root diameters.

Table 3
Root traits (mean ± SD) of different plant species at a depth of 0–30 cm.

Vegetation communities	Soil depth (cm)	RLD_{fine} (cm/cm^3)	RLD_{coarse} (cm/cm^3)	$RSAD_{fine}$ (cm^2/cm^3)	$RSAD_{coarse}$ (cm^2/cm^3)
Artemisia gmelinii	0–10	0.39 ± 0.26b	0.22 ± 0.28a	0.12 ± 0.11a	0.24 ± 0.32a
	10–20	0.19 ± 0.03b	0.04 ± 0.02a	0.06 ± 0.03ab	0.04 ± 0.02a
	20–30	0.16 ± 0.07ab	0.04 ± 0.03a	0.04 ± 0.00b	0.03 ± 0.04a
Bothriochloa ischaemum	0–10	1.17 ± 0.23a	0.02 ± 0.03a	0.25 ± 0.20a	0.02 ± 0.03a
	10–20	0.75 ± 0.07a	0.00 ± 0.00b	0.10 ± 0.02a	0.00 ± 0.00b
	20–30	0.40 ± 0.23a	0.00 ± 0.00a	0.06 ± 0.02a	0.00 ± 0.00a
Abandoned land	0–10	0.25 ± 0.05b	0.02 ± 0.00a	0.07 ± 0.03a	0.01 ± 0.00a
	10–20	0.03 ± 0.01c	0.04 ± 0.02a	0.03 ± 0.00b	0.04 ± 0.02a
	20–30	0.02 ± 0.00b	0.03 ± 0.01a	0.02 ± 0.00b	0.03 ± 0.01a

RLD_{fine}, root length density of fine roots with a diameter < 2 mm; RLD_{coarse}, root length density of coarse roots with a diameter ≥ 2 mm; $RSAD_{fine}$, root surface area density of fine roots with a diameter < 2 mm; RLD_{coarse}, root surface area density of coarse roots with a diameter ≥ 2 mm.
Different letters indicate significant differences among different vegetation communities at a significance level of $p < 0.05$.

community (Fig. 3b). Among the plant patches, *A. gmelinii* exhibits the highest infiltration rates in the *IIR* (15.65 ± 0.45 mm/min), *SIR* (1.53 ± 0.10 mm/min) and *AIR* (4.17 ± 0.11 mm/min), which are 1.97, 2.22 and 1.83 times those for the abandoned land, respectively (Fig. 3b). Furthermore, significant differences ($p < 0.05$) in infiltration rates are observed between the *A. gmelinii* and abandoned land but not between the *B. ischaemum* and abandoned land (Fig. 3b). Among the bare patches, the *B. ischaemum* shows the highest *IIR* (9.68 ± 3.02 mm/min), while the *A. gmelinii* exhibits the highest *SIR* (1.50 ± 0.14 mm/min) and *AIR* (2.56 ± 0.38 mm/min) (Fig. 3b). In addition, the infiltration rates of the plant patches are higher than those of bare patches for the same community (Fig. 3b).

The maximum infiltration depth (*MID*) (22.3–34.7 cm) in the plant patches is generally higher than that in the bare patches (17.7–31.8 cm) for each community, except for *B. ischaemum* (Fig. 4). The *A. gmelinii* and *B. ischaemum* communities display higher *MID* than abandoned land for both patches (Fig. 4). Significantly positive correlations are identified between infiltration rates (*IIR*, *SIR*, and *AIR*) and *MID* ($p < 0.05$, Fig. 5).

3.3. Soil infiltration capacity index

The results show that the *SIC* values of the *A. gmelinii* and *B. ischaemum* communities are significantly higher than that of the abandoned land ($p < 0.05$). For each community, the *SIC* of the plant patch is higher than that of bare patch (Fig. 6a). Moreover, a significantly positive correlation is observed between the *SIC* and *MID* (Fig. 6b), and the correlation coefficient ($R^2 = 0.48$, $p < 0.01$) is higher than that between the *MID* and *SIS* ($R^2 = 0.28$, $p < 0.05$) or *AIR* ($R^2 = 0.44$, $p < 0.01$).

3.4. Relationships between root traits, soil properties and infiltration rates

The *RMD* is positively correlated with the soil infiltration rates (*IIR*, *SIR*, and *AIR*) and *SIC* ($p < 0.01$, Fig. 7). The root traits of the fine roots (RLD_{fine10} and $RSAD_{fine10}$) significantly and negatively correlated with the soil infiltration rates (*IIR*, *SIR*, and *AIR*) and the *SIC* ($p < 0.05$), while the root traits of the coarse roots ($RLD_{coarse20}$ and $RSAD_{coare20}$) had a positive effect on the *IIR* ($p < 0.05$, Fig. 8).

These results show that the *MWD* (0–10 cm and 10–20 cm) is strongly correlated with the *IIR*, *AIR*, and *SIC* ($p < 0.05$), whereas there is nonsignificant correlation between the *TP* (0–30 cm) and the soil infiltration rates (*IIR*, *SIR*, and *AIR*) and *SIC* ($p > 0.05$, Fig. 8).

To identify the main factors that contribute to soil infiltration capacity, a multiple regression was applied to test the various factors pronouncedly affecting the infiltration rates during the different states (Table 4). Consequently, the MWD_{20}, $RSAD_{fine10}$, and $RSAD_{fine20}$ are suggested to be the main influencing factors. The MWD_{20} is considered to be the dominant positive contributor to the *AIR* (63.9 %). The MWD_{20}, $RSAD_{fine10}$, and $RSAD_{fine20}$ explain 98.6 % of the *IIR*. Similarly, the MWD_{20} and $RSAD_{fine10}$ explain 81.7 % of the *SIR* and 91.3 % of the *SIC*, respectively.

4. Discussion

4.1. Soil infiltration capacity under mosaic pattern in natural revegetation grasslands

Compared with the abandoned land, the communities of *A. gmelinii* and *B. ischaemum* show much higher infiltration rates and *SIC* values, not only in plant patches, but also in bare patches (Figs. 3 and 6a), suggesting that natural grasslands restoration can efficiently improve

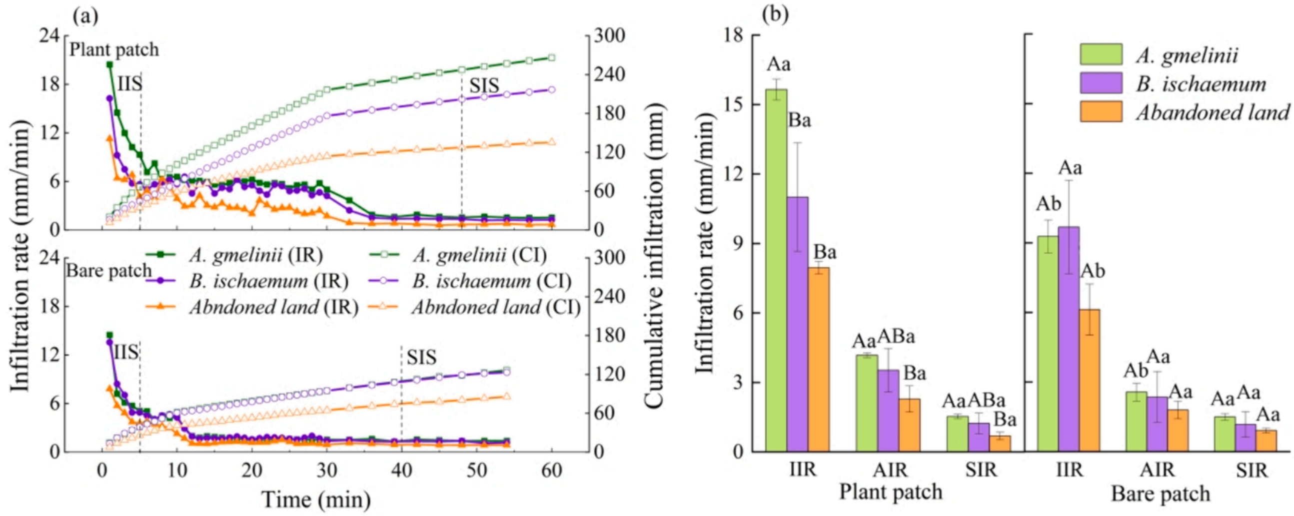

Fig. 3. Variation of soil infiltration over time. (a) Infiltration rate and cumulative infiltration of the plant patches and bare patches for different communities. (b) Infiltration rates at different infiltration states. *IIS*, initial infiltration state (0−3 min); *SIS*, steady infiltration state. *IR*, infiltration rate; *CI*, cumulative infiltration. *IIR*, *SIR*, and *AIR* represent the infiltration rates in the initial, steady, and average infiltration state, respectively. Different capital letters indicate significant differences among the different vegetation communities within the same patch type at a significance level of $p < 0.05$ (Duncan's post hoc tests). Different lowercase letters indicate significant differences between different patches within the same community at a significance level of $p < 0.05$.

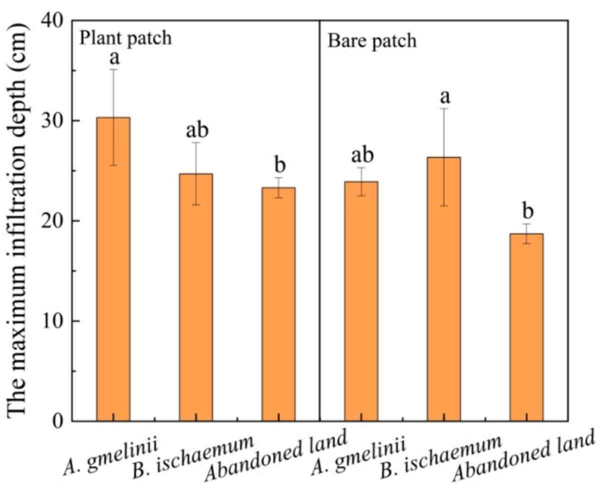

Fig. 4. The maximum infiltration depth (*MID*) for two patches of different vegetation communities. Different letters represent significant differences among the different vegetation communities within the same patch type at a significance level of $p < 0.05$.

soil infiltration capacity (Wang et al., 2013; Zhao et al., 2013). Furthermore, the infiltration rates and *SIC* of the *A. gmelinii* with tap root systems are higher than that of the *B. ischaemum* community with fibrous root systems (Figs. 3 and 6a), which can be attributed to significant differences ($p < 0.05$) in root morphological and distribution traits of plants (Fig. 2 and Table 3). This finding is consistent with the results of previous studies showing that plant species with tap and fibrous root systems have different root morphology and structure traits and differ in soil infiltration capacity in arid and semi-arid regions (Archer et al., 2002; Huang et al., 2016; Liu et al., 2020). Spatial distribution of root channels can be directly determined by root morphological traits, which then regulates the soil water flow path (Hao et al., 2020; Liu et al., 2020). In this study, the roots of *A. gmelinii* with larger-diameter and longer almost-straight taproot can penetrate deeper soil layer, and thus create deeper soil water flow path (Figs. 2, 4 and Table 3).

Plant patches exhibit substantially higher soil infiltration rates (*IIR*, *AIR*, and *SIR*) and *SIC* than bare patches in natural revegetation grasslands (Figs. 3 and 6a), which indicate the redistribution effects of patchy mosaic-pattern on soil water. This finding is consistent with previous studies (Eldridge et al., 2015), which reported substantially greater infiltration under shrub and grass patches than the corresponding interspaces. Some studies believed that the patchy mosaic pattern, as a self-organization phenomenon, is driven by a positive feedback between local plant growth and overland-water flow toward the growth location induced by differential infiltration in arid and semi-arid ecosystems (Getzin et al., 2016; Li, 2011; Li et al., 2022). The plant patches show higher root densities, *MWD*, and *TP* (Table 2 and 3), promoting soil water infiltration and serving as a "sink". Whereas the water infiltration capacity of bare patches is relatively weak (Figs. 3 and 6a), which act as the "source" of surface runoff and sediment (Li, 2011; Merino-Martín et al., 2012). This result in turn further promotes plant growth in plant patches. Therefore, the vegetation-water feedback associated with infiltration heterogeneity has strong self-regulatory ability to adapt climate change by balancing plant growth and limited water (Getzin et al., 2016; Li et al., 2022). Altogether, these findings strongly support the view that the emergence of patchy mosaic patterns can redistribute water on a small scale, allowing vegetation to grow comparatively steadily in arid and semi-arid regions, promoting water infiltration into deeper soil, and reducing the likelihood of surface erosion.

Preferential flow is one of the main forms of water infiltration into the soil (Zhang et al., 2019). The continuity and connectivity of the preferential flow path network can quickly transport and guide water towards deeper soil layers (Jiang et al., 2019). In this study, the two vegetation communities (*A. gmelinii* and *B. ischaemum*) exhibit the higher *MID* than the abandoned land (Fig. 4), suggesting that natural restoration grasslands promote the formation of preferential flow. In particular, plant patches of *A. gmelinii* with tap root systems show the largest *MID* (Fig. 4), which may be associated with the significantly ($p < 0.05$) larger coarse root (diameter $\geq$ 2 mm) density (Table 3). Coarse roots with high penetration tend to increase preferential flow paths by developing pore size and connectivity, and thus facilitate water toward deeper soil layer (Bodner et al., 2014; Lu et al., 2020). Additionally, plant patches exhibit substantially higher *MID* than bare patches in the same community, except for *B. ischaemum* (Fig. 4), which may be attributed to the dense fine roots in the plant patches of *B. ischaemum* (Fig. 2 and Table 3). The fine roots (diameter < 2 mm) in the

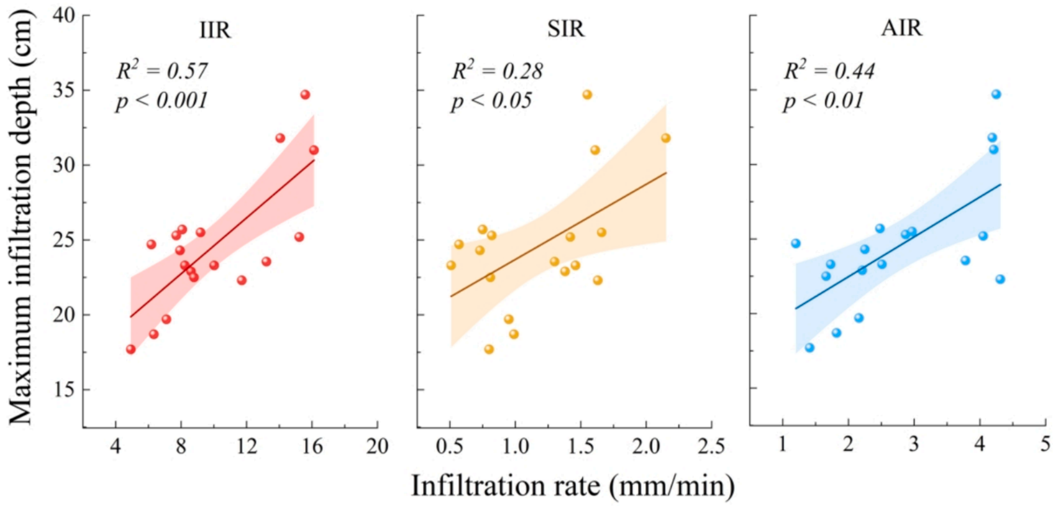

Fig. 5. The relationships between maximum infiltration depth and soil infiltration rates at various infiltration states. *IIR*, *SIR*, and *AIR* represent the infiltration rates in the initial, steady, and average infiltration state, respectively. The shaded areas represent 95% confidence intervals.

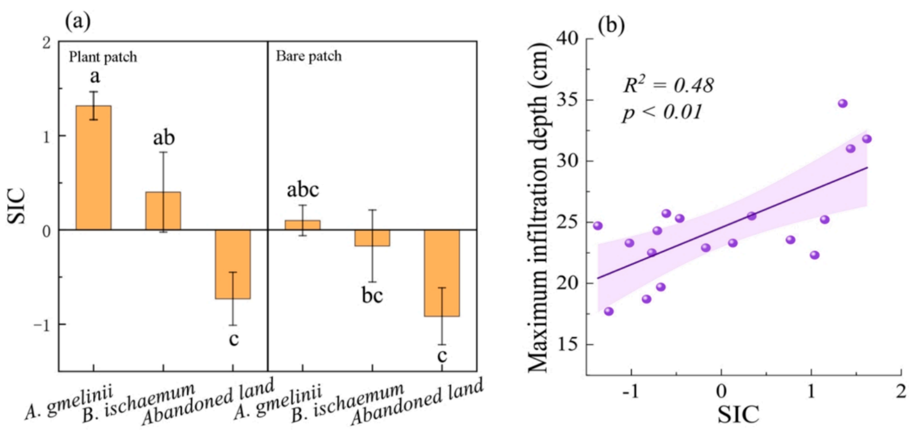

Fig. 6. Soil infiltration capacity index (*SIC*) and its relationship with maximum infiltration depth. (a) *SIC* for different vegetation communities and their patches. (b) The relationship between maximum infiltration depth and *SIC*. Different letters represent significant differences among six different communities and patches at a significance level of $p < 0.05$. The shaded area represents 95 % confidence interval.

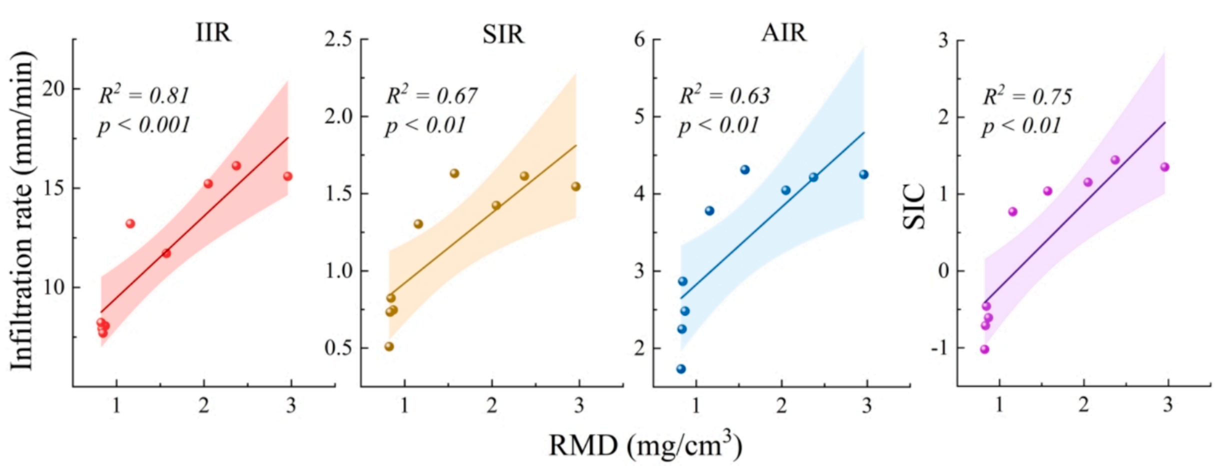

Fig. 7. The relationships between the root mass density (*RMD*) and soil infiltration rates at various infiltration states. *IIR*, *SIR*, and *AIR* represent the infiltration rates in the initial, steady, and average infiltration state, respectively. *SIC* represents soil infiltration capacity index. The shaded areas represent 95% confidence intervals.

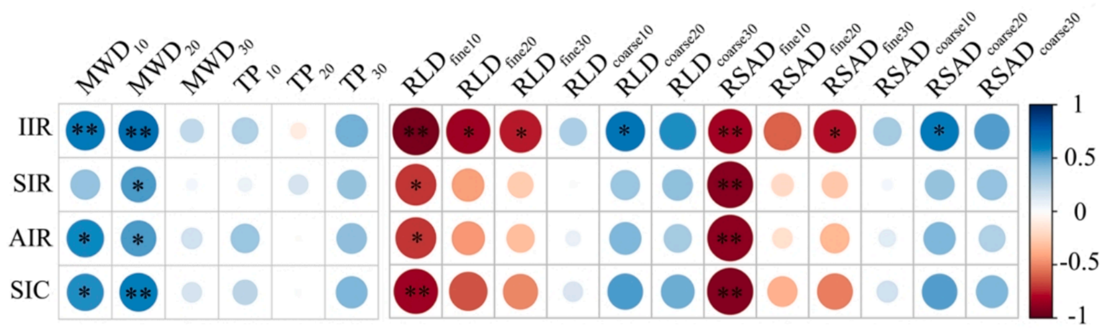

Fig. 8. Pearson correlations for root traits, soil physical properties and infiltration rates. *IIR*, *SIR*, and *AIR* represent the infiltration rates in the initial, steady, and average infiltration state, respectively. *SIC* represents soil infiltration capacity index. *MWD, TP, RLD*, and *RSAD* represent the mean weight diameters of soil aggregates, total porosity, root length density, and root surface area density, respectively. The subscripts 10, 20, and 30 denote the soil depths of 0−10, 10−20, and 20−30 cm, respectively. The subscript fine (coarse) refers to the fine (coarse) roots with diameters < (≥) 2 mm. The blue colour indicates a positive correlation, and the red colour indicates a negative correlation at the following significance levels: $*p < 0.05$; $** p < 0.01$.

Table 4
Multiple-regression equations between the infiltration rates (*IIR*, *SIR*, and *AIR*), soil infiltration capacity index (*SIC*) and root traits, soil physical properties.

	Equation	R^2
IIR	$Y_1 = 7.788 + 1.253MWD_{20} - 0.379RSAD_{fine10} - 0.533RSAD_{fine20}$	0.986
AIR	$Y_2 = 1.834 + 1.299MWD_{20}$	0.639
SIR	$Y_3 = 0.478 + 0.779MWD_{20} + 0.645RSAD_{fine10}$	0.817
SIC	$Y_4 = -0.987 + 1.005MWD_{20} - 0.469RSAD_{fine10}$	0.913

Notes: *IIR*, *SIR*, and *AIR* represent the infiltration rates in the initial, steady, and average infiltration state, respectively. MWD_{20} represents the mean weight diameter of soil aggregates at depths of 10–20 cm; $RSAD_{fine10}$ and $RSAD_{fine20}$ represent the root surface area density of fine roots with diameters < 2 mm at depths of 0–10 cm and 10–20 cm, respectively.

B. ischaemum with fibrous root systems account for 99 % of the total length in this study (Fig. 2b and Table 3), which can form the abundant root network. Although fine roots at low density tend to promote the occurrence of preferential flow, its promotion effect decreases with the increase of root density (Liu et al., 2020). In our study, although *MID* increases in both plant and bare patches of *B. ischaemum* relative to abandoned land, *MID* is lower in plant patches with the higher fine root density compared to bare patches (Figs. 2, 4 and Table 3). In fact, the fine roots at high density are prone to eliminate macropores (Li et al., 2022; Lu et al., 2020), leading to reduced preferential flow channels and lower *MID*. Furthermore, the finding that significantly positive correlations are identified between soil infiltrability (*IIR*, *SIR*, *AIR*, and *SIC*) and *MID* in the vertical profiles ($p < 0.05$, Figs. 5 and 6b), leading to the ineluctable conclusion that the formation of a preferential flow in soil contributes to the enhancement of soil infiltration capacity. Therefore, compared to species with fibrous root systems (e.g., *B. ischaemum*), species with tap root systems (e.g., *A. gmelinii*) are more likely to create effective and stable macropores as preferential flow paths, contributing to the increased infiltration capacity (Guo et al., 2019; Wu et al., 2016).

4.2. Effects of soil and root traits on soil infiltration capacity

Soil properties and root traits affect soil infiltration capacity, which has been widely accept (Leung et al., 2018; Liu et al., 2020; Wu et al., 2016). In this study, higher soil infiltration capacity is associated with higher *MWD* and *TP*, and the crucial role of the root systems (Figs. 7 and 8), corresponding to the results of studies in the adjacent areas and the eastern Loess Plateau (Liu et al., 2020; Tang et al., 2018). The infiltration rates (*IIR*, *SIR*, and *AIR*) and *SIC* are directly affected by the *MWD*, especially for the *MWD* in the 0–20 cm soil layer (Fig. 8). The greater stability of soil aggregates increases the soil porosity and reduces the cover of clay particles in the flow channel (Bronick and Lal, 2005; Murielle et al., 2011), and thus can further improve the permeability in soil. However, the effect of *TP* on soil infiltration capacity is not significant in this study (Fig. 8), which may be attributed to the different effects of fine roots and course roots on soil porosities. The fine roots at high density are prone to occupy and eliminate macropores and have lower macro-porosity, but this is linked to a shift towards fine pore classes rather than a decrease in total porosity (Bodner et al., 2014). Thus, although *B. ischaemum* has higher *TP* with fine roots at high density (Fig. 2b and Table 2), it shows no higher soil infiltration than *A. gmelinii* (Fig. 3). In contrast, coarse roots tend to cause small-scale compaction and the development of macropore, leading to increased saturated hydraulic conductivity and soil infiltration rates (Li et al., 2022; Lu et al., 2020). In addition, root traits have a better relationship with the soil infiltration capacity, with a higher infiltration rate related to a higher *RMD* value (Fig. 7, $p < 0.01$), suggesting that roots are a crucial factor in plant related effects on soil infiltration. This result is consistent with the results of studies in the western Loess Plateau and the middle reaches of the Yangtze River, which showed that soil infiltration capacity is positively correlated with root mass density (Hao et al., 2020; Huang et al., 2016). Actually, roots can release exudates to increase soil organic matter (Bodner et al., 2014; Doerr et al., 2000), enmesh and realign soil particles upon penetration to enhance the stabilization of soil aggregation and the formation of new inter-aggregate pores (Galloway et al., 2018; Six et al., 2004), and determine the distribution of root channels to influence variation of soil porosities (Freschet and Roumet, 2017; Gould et al., 2016), and thus alter the soil water flow channels and paths as well as soil infiltration rates. Although there is a strong positive correlation ($p < 0.01$) between total *RMD* and soil infiltration (Fig. 7), the effects of fine and coarse roots on soil infiltration are different (Fig. 8). The root traits (*RLD* and *RSAD*) of fine roots have a significantly negative effect on soil infiltration rates, while those of coarse roots have an opposite effect (Fig. 8). This disparity could be ascribed to their different turnover rates and physical properties (Archer et al., 2002; Liu et al., 2020; Lu et al., 2020). Coarse roots are characterized by high stiffness, strong penetration, and slow turnover rate, which tends to create new growth paths and grow to deeper soil depths (Bodner et al., 2014; Pierret et al., 2007). Therefore, coarse roots could induce a strong movement and re-orientation of soil particles to increase inter-aggregate pore space, facilitating water permeation through deeper soil depths and improving soil infiltration capacity (Guo et al., 2019; Lu et al., 2020). In contrast, fine roots are characterized by high flexibility, low penetration, and rapid turnover, which are prone to grow along the tortuous pathways using existing pores (growth paths) and form dense root network, resulting in pore heterogenization and induced soil water infiltration by occupying, clogging, or dividing soil macropores (Archer et al., 2002; Fischer et al., 2014; Leung et al., 2015). Additionally, the effects of roots on soil infiltration are related to soil texture (Wang et al., 2023). Relatively large interparticle pores in silty loam with high sand content (0.02−2 mm in diameter) are similar in diameter to fine roots, tending to be blocked by fine roots and thereby causing weakened soil infiltration capacity (Tang et al., 2018). Thus, the *A. gmelinii* with coarser roots exhibits a greater *SIC*. From an ecohydrological perspective, plant species with tap root systems should be more considered in vegetation restoration in the study area and other regions with similar soil texture and hydroclimatic condition.

Based on the multiple regression analysis, it can be concluded that the main factors affecting soil infiltration capacity are MWD_{20}, $RSAD_{fine10}$, and $RSAD_{fine20}$ (Table 4). Furthermore, the interaction effect of soil properties and root traits on soil infiltration rates is much larger than individual effect, which accounts for 48.84 % of the total variance of soil infiltration rates (Fig. 9). These results suggest that rather than working separately, there is a coupled and integrated role for soil and root traits in impacting soil infiltration capacity (Zhang et al., 2023). However, root traits (63.09 %) show relatively more contributions to soil infiltration rates in this study (Fig. 9), which is associated with contrasting root types (Fig. 2b and Table 3). It should not be ignored that different species with various root traits (e.g., *RLD* and *RSAD*) have different effects on soil physical properties, which will result in differences in soil infiltrability (Leung et al., 2015; Neris et al., 2012; Wu et al., 2014). In addition, soil infiltration was determined using the double-ring infiltrometer in this study, a widely accepted method to accurately measure soil infiltration rate curves. Although an optimal time step and insertion depth are taken into consideration, this method is not suitable for assessing the influences of raindrop impact and rain intensity on soil infiltration under natural rainfall conditions (He et al., 2020). Furthermore, the measurement of soil porosity in present study limits the research of the effects of root channels and soil porosities on infiltration paths considering that soil porosity acts as key water flow paths (Guo et al., 2019; Marquart et al., 2020). Further studies are needed to determine the distribution of soil porosity more precisely using computed tomography.

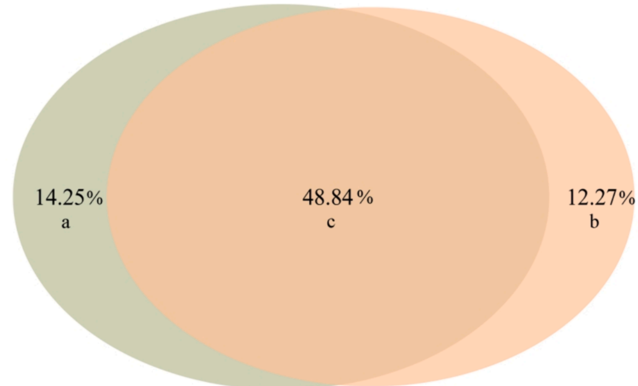

Fig. 9. Variance of soil infiltration rates explained by soil properties and root traits using variation partitioning analysis. a and b indicate the individual effects of root traits and soil properties, respectively. c indicates the interactive effects.

5. Conclusions

In this study, the soil infiltration capacity between different plant communities and contrasting patches was investigated in the loess hilly and gully region, and the main factors affecting the soil infiltration rate were discussed. The results show that natural vegetation restoration can significantly improve the soil infiltration capacity in both plant and bare patches. Moreover, plant patches have greater soil infiltration capacity and promote water infiltration into deeper soil. This result demonstrates that the emergence of patchy mosaic patterns can make infiltration heterogeneity and redistribute water on a slope scale to allow vegetation to grow comparatively steadily in water-limited regions. Compared to the *B. ischaemum* community (fine root: 99 %) with fibrous root systems, the *A. gmelinii* (coarse root: 74 %) with tap root systems, can produce more effective macropores, which act as preferential flow paths, resulting in greater *SICs*. The traits of the fine roots have a significantly negative effect on the soil infiltration rates, while those of the coarse roots positively affect the soil infiltration rate. In addition, the increase in the *MWD* and *TP* improve the soil infiltration capacity. These findings emphasize the important roles of patchy mosaic patterns in the stabilization of ecological systems in arid and semi-arid areas, and improve the understanding of the effect of plant roots on soil water supply.

CRediT authorship contribution statement

Peipei Wang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Qian Liu:** Validation, Methodology, Investigation. **Zhengchao Zhou:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jun'e Liu:** Writing – review & editing, Investigation. **Liguo Cao:** Writing – review & editing, Validation. **Ning Wang:** Validation, Investigation. **Yuying Cao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Financial assistance for this work was supported by the National Natural Science Foundation of China (Grant No. 42277320, 42077058, and 42377322), Natural Science Basic Research Program of Shaanxi (Program No. 2023-JC-JQ-27), Key Research and Development Program of Shaanxi (Program No. 2021ZDLSF05-02), Excellent Graduate Training Program of Shaanxi Normal University (Program No. LHRCTS23079), and Fundamental Research Funds for Central Universities (GK202309005).

References

Aguiar, M.n.R., Sala, O.E., 1999. Patch structure, dynamics and implications for the functioning of arid ecosystems. *Trends in Ecology & Evolution* 14, 273-277.
Anderson, R.L., Brye, K.R., Wood, L.S., 2020. Landuse and soil property effects on infiltration into Alfisols in the lower Mississippi River valley, USA. Geoderma Regional, e00297.
Archer, N.A.L., Quinton, J.N., Hess, T.M., 2002. Below-ground relationships of soil texture, roots and hydraulic conductivity in two-phase mosaic vegetation in South-east Spain. Journal of Arid Environments 52, 535–553.
Bochet, E., Poesen, J., Rubio, J.L., 2006. Runoff and soil loss under individual plants of a semi-arid Mediterranean shrubland: influence of plant morphology and rainfall intensity. Earth Surface Processes and Landforms 31, 536–549.
Bodner, G., Leitner, D., Kaul, H.P., 2014. Coarse and fine root plants affect pore size distributions differently. Plant and Soil 380, 133–151.
Bronick, C.J., Lal, R., 2005. Soil structure and management: A review. Geoderma 124, 3–22.
Cui, Z., Liu, Y., Jia, C., Huang, Z., He, H., Han, F., Shen, W., Wu, G.-L., 2018. Soil water storage compensation potential of herbaceous energy crops in semi-arid region. Field Crops Research 223, 41–47.
Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 2000. Soil water repellency: Its causes, characteristics and hydro-geomorphological significance. Earth-Science Reviews 51, 33–65.
Eldridge, D.J., Wang, L., Ruiz-Colmenero, M., 2015. Shrub encroachment alters the spatial patterns of infiltration. Ecohydrology 8, 83–93.
Fischer, C., Roscher, C., Jensen, B., Eisenhauer, N., Baade, J., Attinger, S., Scheu, S., Weisser, W.W., Schumacher, J., Hildebrandt, A., 2014. How do earthworms, soil texture and plant composition affect infiltration along an experimental plant diversity gradient in grassland? PLOS ONE 9, e98987.

Flury, M., Flühler, H., 1995. Tracer characteristics of brilliant blue FCF. Soil Science Society of America Journal 59, 22–27.
Freschet, G.T., Roumet, C., 2017. Sampling roots to capture plant and soil functions. Functional Ecology 31, 1506–1518.
Galloway, A.F., Pedersen, M.J., Merry, B., Marcus, S.E., Blacker, J., Benning, L.G., Field, K.J., Knox, J.P., 2018. Xyloglucan is released by plants and promotes soil particle aggregation. New Phytologist 217, 1128–1136.
Getzin, S., Yizhaq, H., Bell, B., Erickson, T.E., Postle, A.C., Katra, I., Tzuk, O., Zelnik, Y.R., Wiegand, K., Wiegand, T., Meron, E., 2016. Discovery of fairy circles in Australia supports self-organization theory. Proceedings of the National Academy of Sciences 113, 3551–3556.
Gould, I.J., Quinton, J.N., Weigelt, A., De Deyn, G.B., Bardgett, R.D., 2016. Plant diversity and root traits benefit physical properties key to soil function in grasslands. Ecology Letters 19, 1140–1149.
Gu, C., Mu, X., Gao, P., Zhao, G., Sun, W., Tatarko, J., Tan, X., 2018. Influence of vegetation restoration on soil physical properties in the Loess Plateau, China. Journal of Soils and Sediments 19, 716–728.
Guo, L., Liu, Y., Wu, G.-L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.-Q., Tian, F.-P., He, H., 2019. Preferential water flow: Influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. Journal of Hydrology 578, 124019.
Hao, H., Wei, Y., Cao, D., Guo, Z., Shi, Z., 2020. Vegetation restoration and fine roots promote soil infiltrability in heavy-textured soils. Soil and Tillage Research 198, 104542.
He, Z., Jia, G., Liu, Z., Zhang, Z., Yu, X., Xiao, P., 2020. Field studies on the influence of rainfall intensity, vegetation cover and slope length on soil moisture infiltration on typical watersheds of the Loess Plateau, China. Hydrological Processes 34, 4904–4919.
Huang, Z., Tian, F.P., Wu, G.L., Liu, Y., Dang, Z.Q., 2016. Legume grasslands promote precipitation infiltration better than gramineous grasslands in arid regions. Land Degradation & Development 28, 309–316.
Jia, X., Shao, M., Yu, D., Zhang, Y., Binley, A., 2019. Spatial variations in soil-water carrying capacity of three typical revegetation species on the Loess Plateau, China. Agriculture, Ecosystems & Environment 273, 25–35.
Jiang, X.J., Chen, C., Zhu, X., Zakari, S., Singh, A.K., Zhang, W., Zeng, H., Yuan, Z.-Q., He, C., Yu, S., Liu, W., 2019. Use of dye infiltration experiments and HYDRUS-3D to interpret preferential flow in soil in a rubber-based agroforestry systems in Xishuangbanna, China. Catena 178, 120–131.
Khan, M.R., Koneshloo, M., Knappett, P.S., Ahmed, K.M., Bostick, B.C., Mailloux, B.J., Mozumder, R.H., Zahid, A., Harvey, C.F., van Geen, A., Michael, H.A., 2016. Megacity pumping and preferential flow threaten groundwater quality. Nature Communications 7, 12833.
Leung, A.K., Garg, A., Coo, J.L., Ng, C.W.W., Hau, B.C.H., 2015. Effects of the roots of Cynodon dactylon and Schefflera heptaphylla on water infiltration rate and soil hydraulic conductivity. Hydrological Processes 29, 3342–3354.
Leung, A.K., Boldrin, D., Liang, T., Wu, Z.Y., Kamchoom, V., Bengough, A.G., 2018. Plant age effects on soil infiltration rate during early plant establishment. Géotechnique 68, 646–652.
Li, X., 2011. Mechanism of coupling, response and adaptation between soil, vegetation and hydrology in arid and semiarid regions (in Chinese). Science China Earth Sciences 41, 1721–1730.
Li, Z., Li, X., Zhou, S., Yang, X., Fu, Y., Miao, C., Wang, S., Zhang, G., Wu, X., Yang, C., Deng, Y., 2022. A comprehensive review on coupled processes and mechanisms of soil-vegetation-hydrology, and recent research advances. Science China Earth Sciences.
Liu, J.e., Zhang, X., Zhou, Z., 2019a. Quantifying effects of root systems of planted and natural vegetation on rill detachment and erodibility of a loessial soil. Soil and Tillage Research 195.
Liu, Y., Fu, B., Lü, Y., Gao, G., Wang, S., Zhou, J., 2013. Linking vegetation cover patterns to hydrological responses using two process-based pattern indices at the plot scale. Science China Earth Sciences 56, 1888–1898.
Liu, Y., Guo, L., Huang, Z., López-Vicente, M., Wu, G.-L., 2020. Root morphological characteristics and soil water infiltration capacity in semi-arid artificial grassland soils. Agricultural Water Management 235, 106153.
Liu, Y., Miao, H.T., Chang, X., Wu, G.L., 2019b. Higher species diversity improves soil water infiltration capacity by increasing soil organic matter content in semiarid grasslands. Land Degradation & Development 30, 1599–1606.
Lozano-Baez, S.E., Cooper, M., Meli, P., Ferraz, S.F.B., Rodrigues, R.R., Sauer, T.J., 2019. Land restoration by tree planting in the tropics and subtropics improves soil infiltration, but some critical gaps still hinder conclusive results. Forest Ecology and Management 444, 89–95.
Lu, J., Zhang, Q., Werner, A.D., Li, Y., Jiang, S., Tan, Z., 2020. Root-induced changes of soil hydraulic properties – A review. Journal of Hydrology 589, 125203.
Marquart, A., Eldridge, D.J., Geissler, K., Lobas, C., Blaum, N., 2020. Interconnected effects of shrubs, invertebrate-derived macropores and soil texture on water infiltration in a semi-arid savanna rangeland. Land Degradation & Development 31, 2307–2318.
Merino-Martín, L., Breshears, D.D., Moreno-de las Heras, M., Villegas, J.C., Pérez-Domingo, S., Espigares, T., Nicolau, J.M., 2012. Ecohydrological Source-Sink Interrelationships between Vegetation Patches and Soil Hydrological Properties along a Disturbance Gradient Reveal a Restoration Threshold. Restoration Ecology 20, 360-368.
Molaeinasab, A., Bashari, H., Mosaddeghi, M.R., Tarkesh Esfahani, M., 2021. Effects of different vegetation patches on soil functionality in the central iranian arid zone. Journal of Soil Science and Plant Nutrition 21, 1112–1124.
Murielle, G., Roy, C.S., Alexia, S., 2011. The influence of plant root systems on subsurface flow: Implications for slope stability. BioScience 61, 869–879.
Neris, J., Jiménez, C., Fuentes, J., Morillas, G., Tejedor, M., 2012. Vegetation and land-use effects on soil properties and water infiltration of Andisols in Tenerife (Canary Islands, Spain). CATENA 98, 55–62.
Pierret, A., Doussan, C., Capowiez, Y., Bastardie, F.o., Pagès, L.c., 2007. Root Functional Architecture: A Framework for Modeling the Interplay between Roots and SoilAll rights reserved. No part of this periodical may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or any information storage and retrieval system, without permission in writing from the publisher. Vadose Zone Journal 6, 269-281.
Qiao, L., Zuo, Z., Zhang, R., Piao, S., Xiao, D., Zhang, K., 2023. Soil moisture–atmosphere coupling accelerates global warming. Nature Communications 14, 4908.
Reynolds, B., Reddy, K.J., 2012. Infiltration rates in reclaimed surface coal mines. Water, Air, & Soil Pollution 223, 5941–5958.
Six, J., Bossuyt, H., Degryze, S., Denef, K., 2004. A history of research on the link between (micro)aggregates, soil biota, and soil organic matter dynamics. Soil and Tillage Research 79, 7–31.
Sun, D., Yang, H., Guan, D., Yang, M., Wu, J., Yuan, F., Jin, C., Wang, A., Zhang, Y., 2018. The effects of land use change on soil infiltration capacity in China: A meta-analysis. Science of the Total Environment 626, 1394–1401.
Tang, B., Jiao, J., Yan, F., Li, H., 2018. Variations in soil infiltration capacity after vegetation restoration in the hilly and gully regions of the Loess Plateau, China. Journal of Soils and Sediments 19, 1456–1466.
Wang, P., Su, X., Zhou, Z., Wang, N., Liu, J.e., Zhu, B., 2023. Differential effects of soil texture and root traits on the spatial variability of soil infiltrability under natural revegetation in the Loess Plateau of China. CATENA 220, 106693.
Wang, S., Fu, B., Gao, G., Liu, Y., Zhou, J., 2013. Responses of soil moisture in different land cover types to rainfall events in a re-vegetation catchment area of the Loess Plateau, China. Catena 101, 122–128.
Wang, H., Zhang, G.-H., Li, N.-N., Zhang, B.-J., Yang, H.-Y., 2019. Variation in soil erodibility under five typical land uses in a small watershed on the Loess Plateau, China. CATENA 174, 24–35.
Wu, G.-L., Zhang, Z.-N., Wang, D., Shi, Z.-H., Zhu, Y.-J., 2014. Interactions of soil water content heterogeneity and species diversity patterns in semi-arid steppes on the Loess Plateau of China. Journal of Hydrology 519, 1362–1367.
Wu, G.-L., Yang, Z., Cui, Z., Liu, Y., Fang, N.-F., Shi, Z.-H., 2016. Mixed artificial grasslands with more roots improved mine soil infiltration capacity. Journal of Hydrology 535, 54–60.
Wu, G.-L., Liu, Y., Yang, Z., Cui, Z., Deng, L., Chang, X.-F., Shi, Z.-H., 2017. Root channels to indicate the increase in soil matrix water infiltration capacity of arid reclaimed mine soils. Journal of Hydrology 546, 133–139.
Wu, G.-L., Cui, Z., Huang, Z., 2021. Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. Geoderma 404, 115289.
Ye, C., Guo, Z., Li, Z., Cai, C., 2017. The effect of Bahiagrass roots on soil erosion resistance of Aquults in subtropical China. Geomorphology 285, 82–93.
Zhang, G., Hu, J., 2019. Effects of patchy distributed Artemisia capillaris on overland flow hydrodynamic characteristics. International Soil and Water Conservation Research 7, 81–88.
Zhang, J., Lei, T., Qu, L., Chen, P., Gao, X., Chen, C., Yuan, L., Zhang, M., Su, G., 2017. Method to measure soil matrix infiltration in forest soil. Journal of Hydrology 552, 241–248.
Zhang, J., Lei, T., Qu, L., Zhang, M., Chen, P., Gao, X., Chen, C., Yuan, L., 2019. Method to quantitatively partition the temporal preferential flow and matrix infiltration in forest soil. Geoderma 347, 150–159.
Zhang, W., Zhu, X., Xiong, X., Wu, T., Zhou, S., Lie, Z., Jiang, X., Liu, J., 2023. Changes in soil infiltration and water flow paths: Insights from subtropical forest succession sequence. Catena 221, 106748.
Zhao, Y., Wu, P., Zhao, S., Feng, H., 2013. Variation of soil infiltrability across a 79-year chronosequence of naturally restored grassland on the Loess Plateau, China. Journal of Hydrology 504, 94–103.
Zhou, Z.C., Shangguan, Z.P., 2007. The effects of ryegrass roots and shoots on loess erosion under simulated rainfall. Catena 70, 350–355.